

Neural Language Models (LMs)

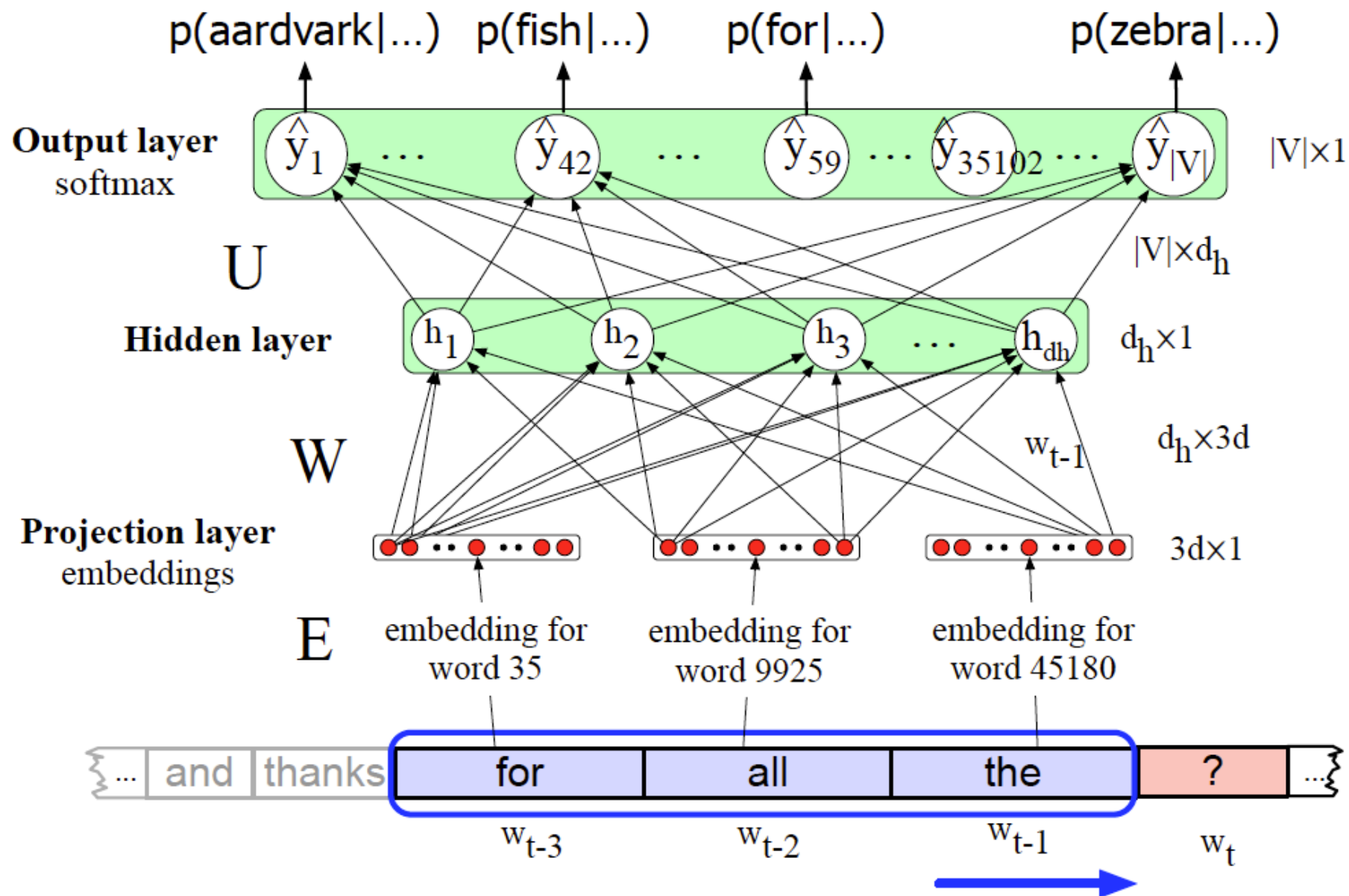
- Recall language modeling
- We've seen n-gram based LMs
- Neural network LMs far outperform n-gram language models
- State-of-the-art neural LMs are based on more powerful neural network technology like Transformers
 - We will talk about them later
- But **simple feedforward LMs** can do well too

Simple FF Neural Language Models

- **Task:** predict next word w_t
- given prior words $w_{t-1}, w_{t-2}, w_{t-3}, \dots$
- **Problem:** Now we're dealing with sequences of arbitrary length
- **Solution:** Sliding windows (of fixed length)

$$P(w_t | w_{t-1}, \dots, w_1) \approx P(w_t | w_{t-1}, \dots, w_{t-N+1})$$

Neural Language Model



Generalization in Neural Language Models

- **Training data:**

- We've seen: I have to make sure that the cat gets fed.
- Never seen: dog gets fed

- **Test data:**

- I forgot to make sure that the dog gets ____
- n-gram LM can't predict "fed"!
- Neural LM will implicitly learn the similarity of "cat" and "dog" embeddings to generalize and predict "fed" after dog